LECTURE 1.4 · SHIP AND OCEAN SYSTEMS DESIGN

Sizing from Regressions: Deadweight and Volume Carriers

Papanikolaou (2026) §1.3.6–1.3.7, §2.1–2.3, Appendix A · Session W1.3 · Lecture 40 min + Exercise 1.4 25 min, partially solved

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From the Owner’s Requirement to the First Dimensions

1

Two routes

- **Deadweight carriers** (tankers, bulk): from the DWT
- **Volume carriers** (containers): from the TEU or hold volume

2

Deadweight route

Regression curves of built ships (book appendix) give Δ, L, B, T, D, CB and lightship from the DWT, with 95 % bands

3

Volume route

- Container ships: from the TEU, by design curves
- Others: from the hold volume

Do the regression curves solve the displacement equation, $\Delta = L \cdot B \cdot T \cdot C_B \cdot \rho$?

A simple test. Three box barges (CB = 1), dimensions L × B × T in m, so Δ = 1.025 · L · B · T. Average them, as a regression curve does, then build a barge from the averages.

Barge A	Barge B	Barge C	Average
90 × 17 × 4.2	100 × 15 × 4.8	116 × 13 × 5.1	102 × 15 × 4.7
Δ = 6,590 t	**Δ = 7,380 t**	**Δ = 7,880 t**	**Δ = 7,280 t**

Built from the averages

102 × 15 × 4.7 → Δ = 7,370 t, 1.2 % above the average

Product of averages ≠ average of products: the curves need not close.

Read: Papanikolaou (2026) §1.3.6–1.3.7, §2.1–2.3; Appendix A.1, A.3 and A.4

The route you take depends on what runs out first

Table 1.8	Deadweight carriers	Book · page
0. Δ	From the given DWT	Tab. 2.1 · 90; App. A · 511
1. L	Slenderness $L/\sqrt[3]{\nabla}$	§2.3, 2.4 · 96, 135
2. CB	L and Fn = V/√(gL)	§2.10 · 168
3. B	Ratios L/B, B/T	§2.5–2.6 · 137–154
4. T	Ratios B/T, L/T	§2.5, 2.8 · 137, 160
5. D	Hold volume, L/D	§2.5, 2.7 · 137, 155
6. CM, CP, CWP	From CB or Fn	§2.9–2.12 · 165–191

Table 1.9	Volume carriers	Book · page
1. L	Hold capacity (bale or net), or TEU number	Figs. 2.26–2.30 · 129–133
2. CB	L and Fn	§2.10 · 168
3. B	L/B, or as for L	§2.5–2.6 · 137–154
4. D	B/D, or capacity/(L·B·D)	§2.5, 2.7 · 137, 155
5. WL	WL/(L·B·D), CB	§2.15.2 · 206
6. DWT	Cargo, fuel and supplies	Eq. 2.111 · 206
7. Δ	WL + DWT	Eq. 2.107 · 199
8. T	From Δ, L, B and CB	§2.8, 2.13 · 160, 192
9. CM, CP, CWP	From CB and Fn	§2.9–2.12 · 165–191

Reminder: using this approach, the design will not resolve the displacement equation. Each dimension comes from its own regression, so the displacement and the hull it implies do not quite agree (Lecture 1.5):

$$\Delta \neq \rho L B T C_B$$

Answer, the book's loop (§1.3.7.1): once Δ comes from the DWT (Table 2.1 or the Appendix A regressions, step 1) and the dimensions from Table 1.8 with the Appendix A regressions (step 2), the book continues:

3 · Powering (§2.14) → **4** · Lines + GA sketch, first ∇ (Ch. 4) → **5** · Weights = displaced water (lines) → **6** · Cargo hold volume (§2.17) → **7** · Minimum freeboard (§2.19.2) → **8** · Stability and trim (§2.18) → **9** · Construction cost (Ch. 6) → **10** · Review (↻ repeat)

From Deadweight to a First Δ, by Ship Family

Table 2.1 · Typical DWT/Δ ratios by ship type (selection) · p. 90

Ship type	Size	DWT/Δ
Crude oil tankers	25–500 kt	0.78–0.88
Bulk carriers	20–200 kt	0.74–0.87
General cargo	5–15 kt	0.65–0.80
Container ships	10–165 kt	0.65–0.76
Ro-Ro cargo	up to 16 kt	0.50–0.60
Ro-Pax ferries	L 85–120 m	0.16–0.33
Tugs	0.5–3 MW	0.20–0.40

$$\Delta = \frac{\mathrm{DWT}}{(\mathrm{DWT}/\Delta)}$$

$$\Delta = W_L + \mathrm{DWT}, \qquad W_L = \text{lightship}$$

- Heavier cargo, larger ratio.
- Volume carriers have small ratios and a wide spread: start from Δ only with good data from sister ships.

For container ships Appendix A gives (Fig. A.29):

$$\mathrm{DWT}/\Delta = 0.5608\,\mathrm{DWT}^{0.024}$$

about 0.70–0.72.

Next slide: the Appendix A fits for one ship type give Δ and every main dimension, sized to the DWT, each with a 95% band.

The Appendix A: Fits for Deadweight Carriers

Quantity	Figs. tank · bulk	Tankers (A.4)	Bulk carriers (A.1)
DWT/Δ	A.51 · A.1	$\mathrm{DWT}/\Delta = 0.5412\,\mathrm{DWT}^{0.0390}$	$\mathrm{DWT}/\Delta = 0.4668\,\mathrm{DWT}^{0.0530}$
Δ	A.52 · A.2	$\Delta = 4774 + 1.1213\,\mathrm{DWT}$	$\Delta = 2.214\,\mathrm{DWT}^{0.9439}$
L (LBP)	A.53 · A.3	$L = 6.864\,\mathrm{DWT}^{0.3072}$	$L = 7.603\,\mathrm{DWT}^{0.3002}$
B	A.54 · A.4	$B = 0.9624\,\mathrm{DWT}^{0.3249}$	$B = 1.0559\,\mathrm{DWT}^{0.3097}$
D	A.55 · A.5	$D = 0.4591\,\mathrm{DWT}^{0.3306}$	$D = 0.5843\,\mathrm{DWT}^{0.3108}$
T	A.56 · A.6	$T = 0.4501\,\mathrm{DWT}^{0.3031}$	$T = 0.4807\,\mathrm{DWT}^{0.2983}$
CB	Tab. 2.15 · A.10	$C_B = 1.06 - 1.68\,F_n$	$C_B = 0.5158\,\mathrm{DWT}^{0.0426}$
Lightship	A.58 · A.9	$W_L = 2.919\,\mathrm{DWT}^{0.7555}$	$W_L/\Delta = 0.6564 - 0.04492\,\ln \mathrm{DWT}$

DWT in t, dimensions in m. Tankers have no CB fit: take it from the speed, Horn (Table 2.15, p. 175), as the course does for the reference ship.

Why we need them

At the start only DWT is known: these fits give a whole first ship in minutes, in line with the fleet.

Then they bound and check

The 95% bands bound the optimization (D6) and check your ship model against real ships (D5).

What they are not

A design method: each fit stands alone, so the set must still be made consistent (Lecture 1.5).

The Appendix A: Fits for Volume Carriers

Quantity	Figs. cont. · reefer	Container ships (A.3) · on DWT	Reefers (A.12) · on hold volume V_{REF}
Start	brief · A.121	DWT ≈ 10 t per TEU (brief)	$\mathrm{DWT} = 1.044\, V_{\mathrm{REF}}^{0.9664}$
Δ	A.30 · A.117	$\Delta = 1.780\,\mathrm{DWT}^{0.9756}$	$\Delta = 1.326\,\mathrm{DWT}^{1.019}$
L (LBP)	A.31 · A.118	$L = 3.541\,\mathrm{DWT}^{0.3884}$	$L = 38.90\,\ln V_{\mathrm{REF}} - 231.3$
B	A.32 · A.119	$B = 1.552\,\mathrm{DWT}^{0.2844}$	$B = 0.8443\,V_{\mathrm{REF}}^{0.3416}$
D	A.33 · A.120	$D = 0.2994\,\mathrm{DWT}^{0.3890}$	$D = 0.4217\,V_{\mathrm{REF}}^{0.3579}$
T	A.34 · A.123	$T = 2.723\,\ln\mathrm{DWT} - 17.16$	$T = 0.1281\,L^{0.8542}$
CB	A.38 · Tab. 2.15	$C_B = 0.9087\,\mathrm{DWT}^{-0.03204}$	$C_B = 1.06 - 1.68\,F_n$ (Horn)
L·B·D	A.35 · —	$L\,B\,D = 1.649\,\mathrm{DWT}^{1.062}$	V_{REF} is the input
Lightship	A.36 · A.117	$W_L = 0.9731\,\mathrm{DWT}^{0.9124}$	$W_L = \Delta - \mathrm{DWT}$

DWT in t, dimensions in m, V_{REF} in m³. Kristensen's curves on the TEU (Figs. A.47–A.50) carry no formula: read them off the graphs (slide 9).

Containers: TEU first, then DWT

The brief gives tonnes per TEU, so the DWT is known and every fit follows. Check the hull volume with L·B·D.

Reefers: the volume itself

The only Appendix A fits whose input is a volume: L, B, D and DWT straight from V_{REF}.

Same caution

Each fit stands alone, and the reefer T comes from L, a fit on a fit. Make the set consistent (Lecture 1.5).

Open Appendix A (GitHub) ↗

Reading Appendix A: a Value and a Band

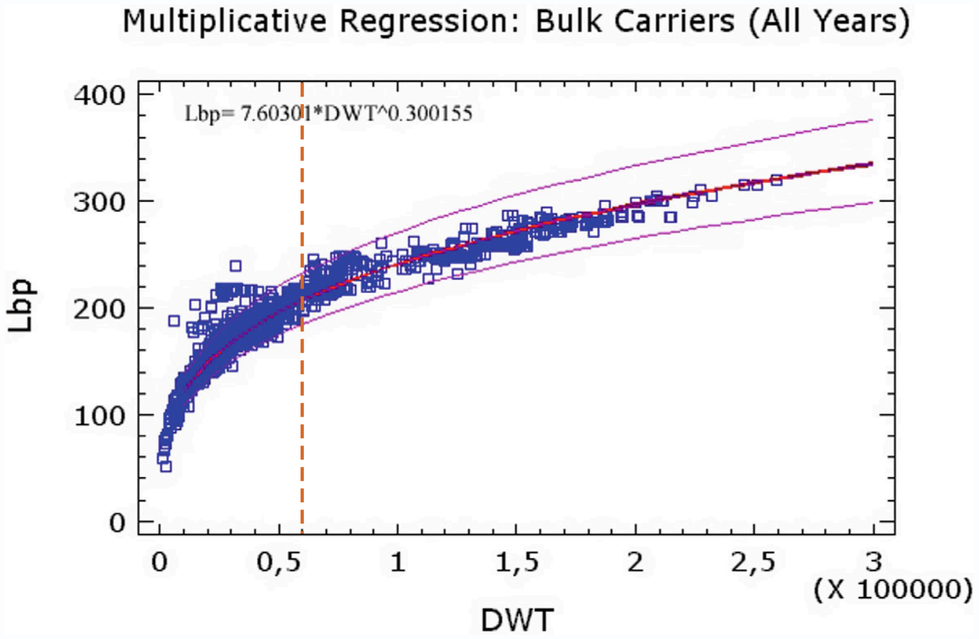


Fig. A.3 · Bulk carriers, LBP against DWT (p. 513) · dashed: 60,000 t

How to read a figure marked *

Red line · the average ship
The mean value at a given DWT.

Purple curves · where real ships are
95 % of real ships lie between them. **Your ship should be inside.**

At 60,000 t: L = 206.7 m on the line; band ≈ 184–230 m, about ±11 %.

The band

The mean is a start, not an answer: **95% of real ships lie inside the band**. A design outside it needs a reason.
The worked examples that follow read every fit this way.

[Open Appendix A (GitHub) ↗]

A 60,000 t Bulk Carrier, Step by Step

Given: a bulk carrier of 60,000 t DWT at 14 kn. Each value below comes from putting DWT = 60,000 into one bulk-carrier formula of slide 5.

1 **Δ from DWT** · the ratio DWT/Δ, Fig. A.1 — band ≈ 69,600–73,900 t

$$\frac{\mathrm{DWT}}{\Delta} = 0.4668 \times 60{,}000^{0.0530} = 0.8363 \quad \Rightarrow \quad \Delta = \frac{60{,}000}{0.8363} = 71{,}740\ \mathrm{t}$$

2 **Length** · LBP fit on DWT, Fig. A.3 — band ≈ 184–230 m

$$L = 7.603 \times 60{,}000^{0.3002} = 206.7\ \mathrm{m}$$

3 **Beam and draught** · Figs. A.4 and A.6 — bands ≈ B 28.4–35.4 m · T 11.4–14.2 m

$$B = 1.0559 \times 60{,}000^{0.3097} = 31.87\ \mathrm{m}, \qquad T = 0.4807 \times 60{,}000^{0.2983} = 12.80\ \mathrm{m}$$

4 **Depth and CB** · Figs. A.5 and A.10 — bands ≈ D 15.9–19.8 m · CB 0.78–0.87

$$D = 0.5843 \times 60{,}000^{0.3108} = 17.85\ \mathrm{m}, \qquad C_B = 0.5158 \times 60{,}000^{0.0426} = 0.824$$

5 **Lightship** · the ratio LS/Δ, Fig. A.9 — band ≈ 9,300–13,600 t

$$\frac{W_L}{\Delta} = 0.6564 - 0.04492 \ln 60{,}000 = 0.162 \quad \Rightarrow \quad W_L = 0.162 \times 71{,}740 = 11{,}640\ \mathrm{t}$$

Seven fits, each with its own **band**: together they **need not balance**.

Does it add up? Almost. Buoyancy ρ·CB·L·B·T = 71,220 t against Δ = 71,740 t (−0.7 %); lightship 11,640 t against Δ − DWT = 11,740 t (−0.9 %).

Fig. A.1 · DWT/Δ

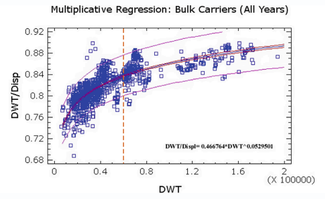


Fig. A.3 · LBP

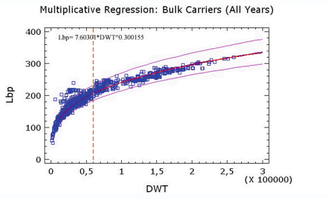


Fig. A.4 · B

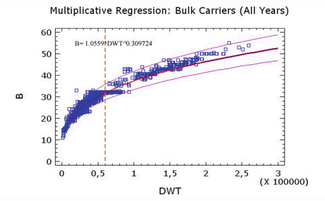


Fig. A.5 · D

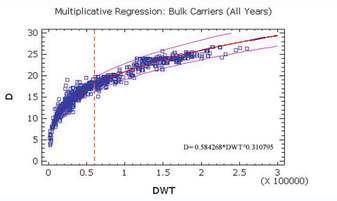


Fig. A.6 · T

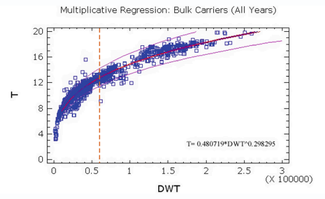


Fig. A.9 · lightship/Δ

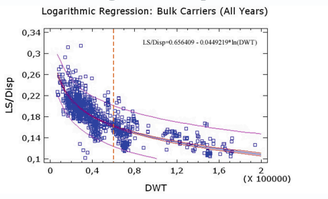
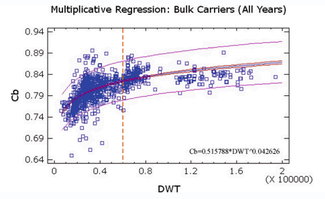


Fig. A.10 · CB



Figs. A.1, A.3–A.6, A.9, A.10 · dashed: DWT 60,000 t

Open Appendix A (GitHub) ↗

A 2,000 TEU Container Ship, Step by Step

Given: a container ship of 2,000 TEU. **Find:** L, B, T, DWT, speed, Δ and CB, from curves of built container ships.

Fig. A.47 · TEU against Lpp

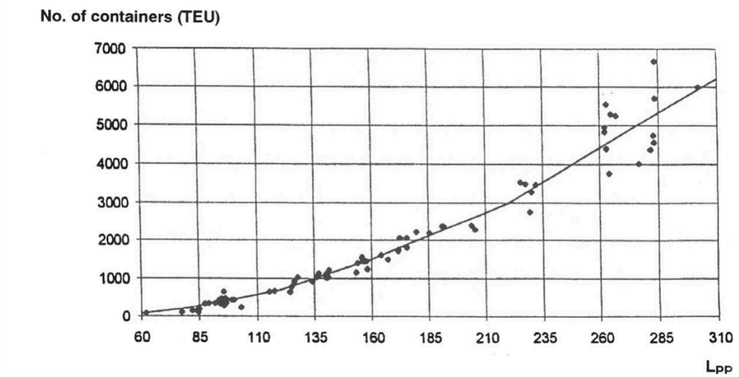
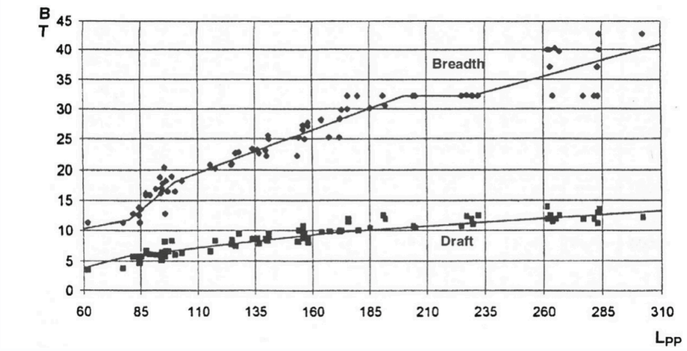


Fig. A.48 · B and T against Lpp



Figs. A.47–A.50 (Kristensen): curves, no printed formula. Fig. A.29: Appendix A fit.

Fig. A.49 · payload

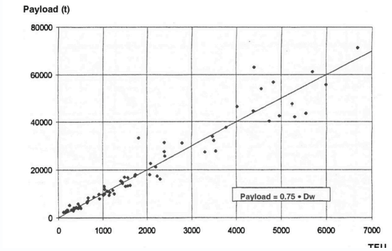


Fig. A.50 · speed

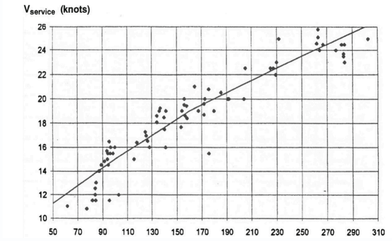
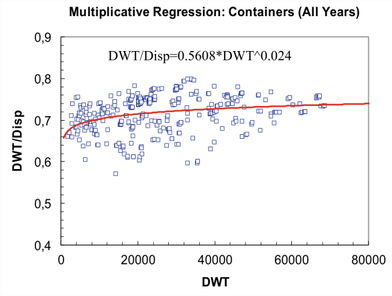


Fig. A.29 · DWT/Δ



1 **TEU → L** · Fig. A.47
2,000 TEU → L ≈ 182 m

2 **L → B and T** · Fig. A.48
B ≈ 30 m · T ≈ 10.2 m

3 **TEU → payload → DWT** · Fig. A.49

$$\mathrm{DWT} = \frac{\text{payload}}{0.75} = \frac{20{,}000}{0.75} = 26{,}700\ \mathrm{t}$$

4 **L → service speed** · Fig. A.50
V ≈ 20.3 kn · Fn = 0.25

5 **Δ from DWT** · Fig. A.29

$$\frac{\mathrm{DWT}}{\Delta} = 0.5608 \times 26{,}700^{0.024} = 0.716 \quad \Rightarrow \quad \Delta = 37{,}230\ \mathrm{t}$$

6 **CB that results** · check it against Horn

$$C_B = \frac{\Delta}{\rho L B T} = \frac{37{,}230}{1.025 \times 182 \times 30 \times 10.2} = 0.65$$

Horn: $C_B = 1.06 - 1.68\,F_n = 1.06 - 1.68 \times 0.247 = 0.645$

Table 2.15. Step 6 gave 0.65: the set is consistent.

Three Checks Before You Trust the Numbers

1

Does the set close?

Each fit stands alone. Compare ρ·CB·L·B·T with Δ, and Δ − DWT with the lightship fit. Bulk carrier: −0.7% and −0.9%.

2

Are the ratios sound?

L/B, B/T, L/D and Fn at the service speed against similar ships. Bulk carrier: 6.49, 2.49, 11.6; Fn = 0.16 at 14 kn.

3

Is the ship inside the data?

A fit outside its data still gives a number, but not an answer. Outside? Change the source: another family, Kristensen’s TEU fits, your D2 ships.

Regressions alone do not close the ship

Each fit is made on its own, so buoyancy and weight need not agree: 0.7 % here, more when CB comes from another source. Exercise 1.4 measures the gap for the reference ship; Lecture 1.5 closes it with the design equation.

Open Appendix A (GitHub) ↗

The Reference Ship from the Tanker Fits

Try first · 5 min, alone or with a neighbour. Reference ship (0.6): DWT 37,600 t, 14.5 kn, draught at most 11.5 m. From the tanker fits (slide 5), estimate Δ, L, B, T, D, CB and the lightship. Does the set close? Does it meet the brief?

1 **Δ from DWT** · the ratio DWT/Δ, Fig. A.51; check with Fig. A.52

$$\frac{\mathrm{DWT}}{\Delta} = 0.541\,\mathrm{DWT}^{\,0.039} = 0.817 \quad \Rightarrow \quad \Delta = \frac{37{,}600}{0.817} = 46{,}050\ \mathrm{t}$$

Fig. A.52: $\Delta = 4774 + 1.1213 \times 37{,}600 = 46{,}935\ \mathrm{t}\ \ (+1.9\,\%)$

2 **L, B, T, D** · one fit each on DWT, Figs. A.53–A.56

$$L = 6.864\,\mathrm{DWT}^{0.3072} = 174.7\ \mathrm{m}, \qquad B = 0.9624\,\mathrm{DWT}^{0.3249} = 29.50\ \mathrm{m}$$
$$T = 0.4501\,\mathrm{DWT}^{0.3031} = 10.97\ \mathrm{m}, \qquad D = 0.4591\,\mathrm{DWT}^{0.3306} = 14.95\ \mathrm{m}$$

3 **CB at 14.5 kn** · Froude number, then Horn (Table 2.15)

$$F_n = \frac{V}{\sqrt{g\,L}} = \frac{14.5 \times 0.5144}{\sqrt{9.81 \times 174.7}} = 0.180 \quad \Rightarrow \quad C_B = 1.06 - 1.68\,F_n = 0.757$$

4 **Lightship** · its own fit on DWT, Fig. A.58; compare with Δ − DWT

$$W_L = 2.919 \times 37{,}600^{0.7555} = 8{,}350\ \mathrm{t} \ \ \text{vs} \ \ \Delta - \mathrm{DWT} = 8{,}450\ \mathrm{t}\ \ (-1.2\,\%)$$

5 **Does the set close?** · buoyancy ρ·CB·L·B·T against Δ

$$\rho\,C_B\,L\,B\,T = 1.025 \times 0.757 \times 174.7 \times 29.50 \times 10.97 = 43{,}880\ \mathrm{t}\ \ (-4.7\,\%)$$

6 **Ratios, Fn, draught** · against similar ships and the 11.5 m limit

$$\frac{L}{B} = 5.92, \quad \frac{B}{T} = 2.69, \quad \frac{L}{D} = 11.7, \quad F_n = 0.180, \quad T = 10.97\ \mathrm{m} < 11.5\ \mathrm{m}$$

Fig. A.51 · DWT/Δ

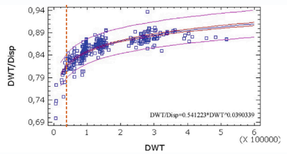


Fig. A.52 · Δ

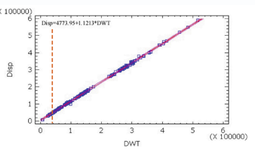


Fig. A.53 · LBP

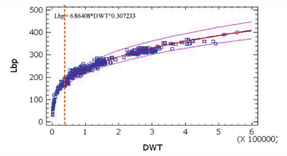


Fig. A.54 · B

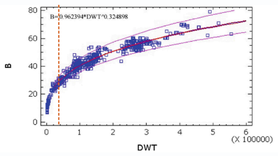


Fig. A.55 · D

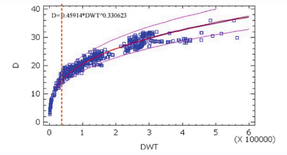


Fig. A.56 · T

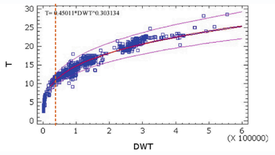
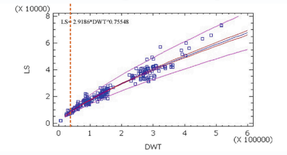


Fig. A.58 · lightship



Figs. A.51–A.56, A.58 · dashed: DWT 37,600 t

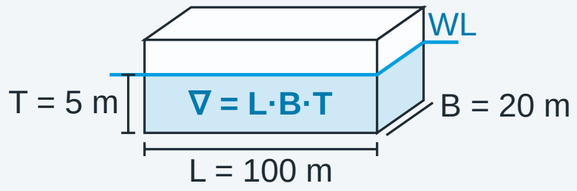
The fits give a plausible ship, but **buoyancy misses Δ by 4.7 %**: slides 12–13 close it by hand, Lecture 1.5 by the design equation. **Your project:** repeat this for your ship (D2) as an initial approach!!

Scale to a Target Volume Without Distortion

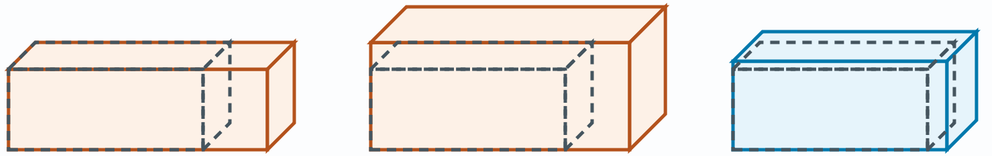
The tempting move: buoyancy is 4.5 % short of the weight (43,880 vs 45,949 t), so why not simply multiply L, B and T by W/Δ?

Test it on a box barge ($C_B = 1$) floating at draught T. Only the **submerged box** counts: ∇ = 10,000 m³, Δ = ρ∇ = 10,250 t. Target W = 13,643 t:

$$W/\Delta = 13\,643/10\,250 = 1.331$$



Submerged box after scaling (dashed: the barge's)



a · stretched **b · far too big** **c · same shape**

Factors (k_L, k_B, k_T)	Submerged L × B × T (m)	Δ′ = $k_L k_B k_T$ · Δ (t)	L/B · B/T	Δ′ = W?	Same shape?
a · (1.331, 1, 1)	133.1 × 20 × 5	1.331 × 10,250 = 13,643	6.66 · 4.00	✓	×
b · (1.331, 1.331, 1.331)	133.1 × 26.62 × 6.655	1.331^3 × 10,250 = 24,169	5.00 · 4.00	× 1.77 W	✓
c · (1.10, 1.10, 1.10)	110 × 22 × 5.5	1.10^3 × 10,250 = 13,643	5.00 · 4.00	✓	✓

Why: each dimension multiplies Δ once

$$\rho\,(k_1 L)(k_2 B)(k_3 T)\,C_B = k_1 k_2 k_3 \cdot \rho\,L\,B\,T\,C_B = k_1 k_2 k_3\,\Delta$$

same shape: $k_1 = k_2 = k_3 = k \;\Rightarrow\; k^3\Delta = W \;\Rightarrow\; k = (W/\Delta)^{1/3}$

Reference ship, round 1: $k = (45\,949/43\,880)^{1/3} = 1.015$ → L, B, T each + 1.5 % (next slide)

Regressions + Design Spiral

Strategy
W fixed

1 · Weight: Δ < W ?
scale L, B, T by $\sqrt[3]{(W/\Delta)}$

2 · Tanks: V < Vreq ? *
deepen D by Vreq/V

3 · Draught: T ≤ 11.5 m ?
check only, here not binding

↻ Repeat
until no check fails

1 · Problem: the owner's brief (0.6)

DWT 37,600 t @ 14.5 kn

Cargo 35,000 t @ 0.90 t/m³, tanks 98 % full

Draught T ≤ 11.5 m (terminal limit)

$$V_{req} = \frac{35\,000}{0.90 \times 0.98} = 39\,680\ \mathrm{m}^3$$

Converged when

$$\frac{|W - \Delta|}{W} \le 1\,\%, \quad V \ge V_{req}, \quad T \le 11.5\ \mathrm{m}$$

2 · Start: Exercise 1.4 (slide 11)

Value	Start	Source, App. A.4
L	174.68 m	Fig. A.53
B	29.50 m	Fig. A.54
T	10.97 m	Fig. A.56
D	14.95 m	Fig. A.55
CB	0.7573	Horn, Fn 0.1802
WL	8,349 t	Fig. A.58, fixed

3 · Checks, simplified: round 1

$$\Delta = \rho L B T C_B = 1.025 \times 174.68 \times 29.50 \times 10.97 \times 0.7573 = 43\,880\ \mathrm{t}$$

Weight = buoyancy, §2.13; ρ = 1.025 t/m³

$$W = W_L + \mathrm{DWT} = 8\,349 + 37\,600 = 45\,949\ \mathrm{t}$$

Lightweight from its regression: fixed, §2.15

$$V = 0.48\,L\,B\,D = 0.48 \times 177.4 \times 29.96 \times 14.95 = 38\,140\ \mathrm{m}^3 \ < \ V_{req}$$

Cargo tanks, §2.17 B.1 (see *)

Round	Weight: Δ vs W	Tanks V (m³)	Draught T (m)	Ship after the round (m)
1	43,880 vs 45,949 t × → L, B, T × 1.015	38,140 × → D × 1.041	11.14 ✓	177.4 × 29.96 × 11.14 × 15.56
2	46,100 vs 45,949 t ✓ (CB 0.760 at the new L)	39,700 ✓	11.14 ✓	Converged: no change

* Tank check, simplified to demonstrate the spiral: $V = C \cdot L \cdot B \cdot D$, the cubic-coefficient method of §2.17 (B.1), with $C = V/(L{\cdot}B{\cdot}D) \approx 0.48$ of a similar double-hull product tanker. **§2.17 in full (methods B.2–B.5): Lecture 3.2, Hold Capacity.**

The Story in Four Lines

1 · Two routes. Deadweight carriers start from the DWT, volume carriers from the TEU or hold volume.

2 · Deadweight route. Put the DWT in each Appendix A fit: a value and a 95 % band, about ±11 % on L, B, T, D.

3 · Volume route. Size from the TEU (Kristensen) or the hold volume; the weights follow, T last.

4 · Do they close? Almost, not exactly: −0.7 % for the bulk carrier, −4.7 % for the reference ship. The design spiral closes it by trial and error; Lecture 1.5 by the design equation.

Next, same day (W1.4): the design equation and Normand’s method; Exercise 1.5 on the reference ship. Read §2.13 and App. C.